

# Kowndinya Boyalakuntla

Third-year Ph.D. student in Computer Science | Rutgers University

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## Research interests

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I work on learning and planning for robot manipulation and whole-body control. My current research explores learned world-models and how to turn human motion capture into useful action abstractions for long-horizon planning in humanoid loco-manipulation.

## Education

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### Rutgers University

*Ph.D. in Computer Science*

Advisor: Prof. Abdeslam Boularias

Sep 2024 – present

### Rutgers University

*M.S. in Computer Science; GPA: 3.8/4.0*

Advisor: Prof. Abdeslam Boularias

Thesis: Context-Aware Entity Grounding with Open-Vocabulary 3D Scene Graphs.

2022 – 2024

### Indian Institute of Technology Tirupati

*B.Tech. in Computer Science and Engineering; GPA: 8.42/10.0*

Advisors: Prof. Sridhar Chimalakonda, Prof. Kalidas Yeturu

Jul 2017 – Jul 2021

## Manuscripts under review

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- PL-MPC: Beyond Policy Alignment: Closing the Planning–Learning Loop for Robot Control with Learned World Models**  
Kowndinya Boyalakuntla, Yuhan Liu, Abdeslam Boularias  
*Manuscript, 2026. Under review at ICRA 2027.* [Paper] [Project] [Code]
- TRACE: Plan-Conditioned Imitation for Robust Object Retrieval under Self-Occlusion in Dense Clutter**  
Kowndinya Boyalakuntla, Ajinkya Pawar, Abdeslam Boularias, Jingjin Yu  
*Manuscript, 2026. Under review at ICRA 2027.* [Paper] [Project] [Code] [Data]

## Research experience

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### Rutgers University

*Graduate Student Researcher · New Brunswick, NJ*

Research in robot learning, visual perception, and manipulation with Prof. Abdeslam Boularias. Work includes scene grounding, language-guided rearrangement, dynamic grasping, learned world models, and manipulation under self-occlusion.

Jan 2023 – present

### RISHA Lab, IIT Tirupati

*Undergraduate Student Researcher · Tirupati, India*

Research in empirical software engineering and human–computer interaction, including tools for evaluating software repositories and generating energy-aware mobile applications.

Dec 2019 – May 2022

### University of Waterloo

*Visiting Scholar · Waterloo, Canada*

Improved RepoReaper for large-scale GitHub repository analysis, increasing recall and reducing false positives without relying on GHTorrent.

Jul 2020 – Sep 2020

## Teaching

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### Instructor, Rutgers University

CS440: Introduction to Artificial Intelligence Undergraduate course

Fall 2026

## Peer-reviewed publications

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1. **KARL: Kalman-Filter Assisted Reinforcement Learner for Dynamic Object Tracking and Grasping**  
Kowndinya Boyalakuntla, Abdeslam Boularias, Jingjin Yu  
*IROS, 2025* [\[Paper\]](#) [\[Code\]](#)
2. **DAP: Diffusion-based Affordance Prediction for Multi-modality Storage**  
Haonan Chang, Kowndinya Boyalakuntla, Yuhan Liu, Xinyu Zhang, Liam Schramm, Abdeslam Boularias  
*IROS, 2024* [\[Paper\]](#) [\[Code\]](#)
3. **LGMCTS: Language-Guided Monte-Carlo Tree Search for Executable Semantic Object Rearrangement**  
Haonan Chang, Kai Gao, Kowndinya Boyalakuntla, Alex Lee, Baichuan Huang, Harish Udhaya Kumar, Jingjin Yu, Abdeslam Boularias  
*IROS, 2024* [\[Paper\]](#) [\[Project\]](#) [\[Code\]](#)
4. **Context-Aware Entity Grounding with Open-Vocabulary 3D Scene Graphs**  
Haonan Chang, Kowndinya Boyalakuntla, Shiyang Lu, Siwei Cai, Eric Pu Jing, Shreesh Keskar, Shijie Geng, Adeeb Abbas, Lifeng Zhou, Kostas Bekris, Abdeslam Boularias  
*CoRL, 2023* [\[Paper\]](#) [\[Project\]](#) [\[Code\]](#)
5. **eGEN: an energy-saving modeling language and code generator for location-sensing of mobile apps**  
Kowndinya Boyalakuntla, Marimuthu Chinnakali, Sridhar Chimalakonda, Chandrasekaran Kalicheti  
*ESEC/FSE, 2022* [\[Paper\]](#) [\[Code\]](#)
6. **RepoQuester: A Tool Towards Evaluating GitHub Projects**  
Kowndinya Boyalakuntla, Meiyappan Nagappan, Sridhar Chimalakonda, Nuthan Munaiah  
*ICSME, 2022* [\[Paper\]](#) [\[Code\]](#)
7. **GitQ-towards using badges as visual cues for GitHub projects**  
Akhila Sri Manasa Venigalla, Kowndinya Boyalakuntla, Sridhar Chimalakonda  
*ICPC, 2022* [\[Paper\]](#) [\[Code\]](#)
8. **A Platform for Large Scale Auto Annotation of Scanned Documents Featuring Real-Time Model Building and Model Pooling**  
Prashanth Komuravelli, Kowndinya Boyalakuntla, Vijay Chilaka, Teja Dande, Vidya Rodge, Ramya Velaga, Reena Abasaheb Deshmukh, Kalidas Yeturu  
*CVIP, 2021* [\[Paper\]](#) [\[Code\]](#)

## Honors & awards

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**2022** Project showcase, IInvenTiv All IITs R&D Fair. Research project advised by Prof. Kalidas Yeturu, showcased at IIT Delhi.

**2019** Gold Medal, Inter IIT Technical Meet 8.0. Ashoka's Tech for Change Challenge at IIT Roorkee; high-preparation event among 20 IITs.

## Leadership & mentoring

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**Google Developer Student Clubs, IIT Tirupati**

Feb 2019 – Jul 2020

*Community Lead*

Organized 12 workshops and hackathons for more than 70 students; led 15 mentoring sessions on Flutter and Android.

## Technical skills

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**Programming:** Python, C++

**Robotics and learning:** PyTorch, TensorFlow, OpenCV, PyBullet, Isaac Gym, ROS, Kornia

**Visualization:** Three.js

## Professional service & activities

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**Reviewer:** IROS (2025, 2026); CoRL (2026); IJRR (2026); ICAPS (2026).

**2021** Developed an Android application for IIT Tirupati's convocation.

**2020** Completed 80 hours of National Service Scheme service; developed the IIT Tirupati NSS website and led its maintenance team.